CHITSHIT BAYES

Basic Notes

Features: Variable that influences target variables.

Target: Target variables are the variables to be predicted.

1. Determine Likelihood and Prior
   Likelihood: Assume the distribution of the data.
   a. Normally Distributed (Linear Regression) $[-\infty, \infty]$

   If the target variable is numerical (angka), **except** only 0 and 1 then assume the likelihood is normally distributed.

   $$Y = \alpha + \sum_{i=1}^{n} \beta_i x_{ij}$$

   With likelihood of $Y_i \sim Normal(\alpha + \sum_{i=1}^{n} \beta_i x_{ij}, \sigma^2)$, where X_{ij} being the value for each iteration of j with Prior of $\alpha \sim Normal(0{,}1000)$, $\beta_i \sim Normal(0{,}1000)$, and $\sigma^2 \sim InvGamma(0.1{,}0.1)$.

   b. Bernoulli Distribution (Classification) -> Logistic Regression Type 1

   if the target variable is **only** 0 and 1, without the number of occurrences of the target variable then assume the likelihood is Bernoulli distributed.

   $$logit[Prob(Y_i = 1)] = \alpha + \sum_{i=1}^{n} \beta_i x_{ij}$$

   With likelihood of $Y_i \sim Bern(Prob(Y_i = 1))$, where $Prob(Y_i = 1)$ being the value for each iteration of i with Prior of $\alpha \sim Normal(0{,}1000)$ and $\beta_i \sim Normal(0{,}1000)$.

   c. Binomial Distribution (Classification) -> Logistic Regression Type 2

   If the target variable is only 0 and 1, without the number of occurrences of the target variable then assume the likelihood is Binomial distributed.

   Example: Basketball Free throw, Y is the number of successes throw and N is the number of tries, X is the features.

   $$Y_i \sim Binomial(N_i, p_i)$$

   Where p_i is the true probability of $Y_i = 1$ using the $logit$ function, or can be denoted as

   $$logit(p_i) = \alpha + \sum_{i=1}^{n} \beta_i x_{ij}$$

   With Prior of $\alpha \sim Normal(0{,}1000)$ and $\beta_i \sim Normal(0{,}1000)$.

   d. Beta Distribution (Webe Tak ngerti logicnya)

2. Model Fitting: Look at the Photocopy based on the likelihood.

3. Feature Selection: Use SVSS to use the features with the highest inc_prob.
4. Model Parameter: To create more then 1 model then change thin.
5. Model Validation:
    a. Check Convergence:
        i. Plotting: If the shape is like a caterpillar (kaki seribu) then it's convergent.
        ii. Gelman_diag: Convergent if multivariate score is < 1.1.
    b. Check Sample Size (> 1000): Preferably more than 1000.
    c. Check DIC & WAIC Score: The lower the better, Prioritize WAIC over DIC.
6. Conclusion:
   Conclude the best model based on the DIC & WAIC Score.

   SEMOGA LULUS.